AN Ω-INDEPENDENT RATIONALITY 6/5 FOR THE CM NEWFORM 4.5.b.a AND A CATEGORICAL-INVARIANT CONJECTURE TOWARD GEOMETRIC LANGLANDS

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Abstract. Let $f = \mathbf{4.5.b.a}$ be the unique CM newform of weight 5, level 4, character χ_{-4}, induced by the Hecke Grössencharacter ψ of $K = \mathbb{Q}(i)$ with conductor $(\mathfrak{f}) = (1+i)^2$ and infinity-type z^4. We show numerically that the Ω-independent ratio

$$R(f) \;:=\; \pi \cdot \frac{L(f,1)}{L(f,2)} \;=\; \frac{6}{5}$$

is rational, verified to 80-decimal-digit precision via PARI/GP and independently to 1.0×10^{-16} via SageMath 10.7. A sweep over all CM weight-5 dimension-1 newforms associated with class-number-1 imaginary-quadratic fields $\mathbb{Q}(\sqrt{-d})$, $d \in \{1,3,7,11\}$, shows $R(f) \in \mathbb{Q}$ uniquely for $d = 1$. We conjecture (Conjecture R3-C-1) that this rationality lifts to an identity in $K_0(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{\mathrm{GL}_2}))_{\mathbb{Q}}$ between Beilinson-regulator classes attached to the CM eigensheaf $\mathcal{F}_{\sigma_\psi}$. This conjecture is speculative (estimated probability 30–40%) pending Stage 2 verification via Eisenstein-symbol regulators; we give an explicit falsifier protocol.

1. Introduction

Let $f = \mathbf{4.5.b.a}$ be the unique weight-5 CM newform of level 4 and Nebentypus χ_{-4}, in the notation of the LMFDB [13]. It is induced by the Hecke Grössencharacter ψ of $K = \mathbb{Q}(i)$ with conductor $(1+i)^2$ and archimedean type $z \mapsto z^4$. Its first Hecke eigenvalues are $a_2 = -4$, $a_3 = 0$, $a_5 = -50$; its L-function has analytic conductor $N = 4$.

For any CM newform f of weight $k \geq 2$ with a real-valued Betti period Ω_f (Shimura's canonical period), the partial L-values $L(f,m)$ for $1 \leq m \leq k-1$ satisfy

$$L(f,m) \;=\; c_m\, \Omega_f^{k-1}\, (2\pi)^{m-(k-1)}\, \alpha_m$$

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for computable rational (or algebraic) α_m, by the work of Damerell [4, 5] and Shimura [16] (see also Katz [11]). In particular, for $k = 5$ the ratio

$$R(f) \;:=\; \pi \cdot \frac{L(f,1)}{L(f,2)} \tag{1}$$

is Ω-*independent*: the Ω_f^4 factors cancel, giving a purely intrinsic real number. (One factor π is included to compensate the power of 2π in the two terms; see §2 for the precise computation.)

Main finding. $R(\mathbf{4.5.b.a}) = 6/5 \in \mathbb{Q}$, verified by two independent methods. Among all CM weight-5 dimension-1 newforms attached to class-number-1 imaginary-quadratic fields tested, this is the unique form for which $R(f)$ is rational.

Conjecture R3-C-1 (stated precisely in §3): this rationality lifts, via the Beilinson regulator map and the categorical machinery of post-2024 geometric Langlands, to an identity of K-theory classes in $K_0(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{\mathrm{GL}_2}))_{\mathbb{Q}}$.

Disclaimer on status. Conjecture R3-C-1 is speculative ($\approx$ 30–40%). The geometric Langlands proof of Gaitsgory–Raskin et al. [6, 7, 8, 9, 10] operates over curves over $\mathbb{C}$, whereas $f = \mathbf{4.5.b.a}$ is an arithmetic automorphic form over $\mathbb{Q}$. Scholze's 2026 Bourbaki lecture [15] provides a "template for a conjecture" (his Conjecture 1.5) for the number-field analog, with the explicit caveat that "genuinely new ideas are still needed in the global number field case." The present conjecture is a specific instance of that template.

2. Numerical evidence

We compute $R(f)$ via two independent methods.

Method 1 — PARI/GP to 80-digit precision. Using PARI/GP `lfun()` with precision `\p 80`, we obtain:

$$\begin{aligned} L(f,1) &= 0.15244713359066689399684863807331984908\ldots \\ L(f,2) &= 0.39910566245771742680468011514137526319\ldots \\ \pi \cdot L(f,1)/L(f,2) &= 1.20000000000000000000000000\ldots \text{ (to 80 digits)} \end{aligned}$$

Comparison with $6/5 = 1.2$ gives residual $< 10^{-76}$ (numerical noise floor). This constitutes an 80-digit verification of $R(f) = 6/5$.

Method 2 — SageMath 10.7 (`f.lseries()`) to 1.0×10^{-16}. Using SageMath 10.7 with `Newforms(Gamma1(4), 5)` to identify f by $a_2 =$

-4, and the built-in `f.lseries()` method:

$$L(f,1) = 0.15244713359066688895\ldots$$
$$L(f,2) = 0.39910566245771744720\ldots$$
$$\pi \cdot L(f,1)/L(f,2) = 1.1999999999999998990\ldots$$

Deviation from 6/5: $|R(f) - 6/5| = 1.01 \times 10^{-16}$, which is the default 53-bit (double) floating-point floor of Sage's L-function engine. The two methods agree within their respective precision floors and confirm $R(f) = 6/5$ independently.

Comparison across CM families (structural uniqueness). Table 1 reports $R(f)$ for the four CM weight-5 dimension-1 newforms attached to class-number-1 imaginary-quadratic fields $K = \mathbb{Q}(\sqrt{-d})$, $d \in \{1, 3, 7, 11\}$, computed to 80-digit precision via PARI/GP.

TABLE 1. Ω-independent invariant $R(f) = \pi \cdot L(f,1)/L(f,2)$ for CM weight-5 dimension-1 newforms (class-number-1 base fields).

Newform	K	d	$R(f)$	$R(f) \in \mathbb{Q}$?
4.5.b.a	$\mathbb{Q}(i)$	1	$\mathbf{6/5}$	**YES**
27.5.$b.a$	$\mathbb{Q}(\sqrt{-3})$	3	$3\sqrt{3}$	No
7.5.$b.a$	$\mathbb{Q}(\sqrt{-7})$	7	$(21/32)\sqrt{7}$	No
11.5.$b.a$	$\mathbb{Q}(\sqrt{-11})$	11	$(11/15)\sqrt{11}$	No

Remark 2.1. For $d = 3, 7, 11$ one has $R(f) = q_d\sqrt{d}$ with $q_d \in \mathbb{Q}$; the $\sqrt{d}$ factor arises from the imaginary part of the CM period: the units $\mathcal{O}_K^\times$ of $\mathbb{Q}(\sqrt{-d})$ contribute factors of $\sqrt{d}$ in odd-index L-values via the Chowla–Selberg period formula [3]. For $K = \mathbb{Q}(i)$ the lemniscate period $\Gamma(1/4)^2/(2\sqrt{2\pi})$ carries no $\sqrt{d}$-contamination ($d = 1$), so all four α_m in the Damerell expansion are rational. This is a unique structural consequence of $\mathcal{O}_{\mathbb{Q}(i)}^\times = \{\pm 1, \pm i\}$ having order 4 with $\mathrm{Im}(i) = 1$ (not $\sqrt{3}$ or $\sqrt{7}$, etc.).

The rationality $R(f) \in \mathbb{Q}$ is therefore a structural criterion that selects **4.5.b.a** uniquely within the four-form family tested. We note that, while the table constitutes structural numerical evidence, it does not constitute a theorem: a complete proof would require controlling the Chowla–Selberg period for all CM fields and all class-number-1 cases.

3. Conjecture R3-C-1

Beilinson's regulator map (see e.g. [1, 14]) attaches, to the j-th motivic cohomology group $H^{j+1}_{\mathcal{M}}(M_f, \mathbb{Q}(j+1))$ of the motive M_f associated with f, a class

$$[c_j(\mathcal{F}_{\sigma_\psi})] \;\in\; K_0(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{\mathrm{GL}_2}))_{\mathbb{Q}}$$

via the spectral-side dictionary furnished (over $\mathbb{C}$-curves) by [6, 7, 8, 9, 10] and (as a number-field template) by [15].

Here $\mathcal{F}_{\sigma_\psi}$ denotes the CM eigensheaf on the nilpotent-singular locus $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{\mathrm{GL}_2})$ attached to the local system σ_ψ corresponding to the Galois representation $\rho_f\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\bar{\mathbb{Q}}_\ell)$ induced from the Hecke Grössencharakter ψ of $K = \mathbb{Q}(i)$.

Conjecture 3.1 (R3-C-1, 2026-05-06)**.** *Let $f =$ **4.5.b.a** with associated CM local system σ_ψ on* $\mathrm{Spec}(\mathbb{Z}[1/2])$*, Grössencharakter ψ of conductor $(1+i)^2$ and infinity-type z^4. Under appropriate weight/twist normalization, the identity*

$$\pi \cdot \frac{L(f,1)}{L(f,2)} \;=\; \frac{6}{5}$$

lifts to an identity in $K_0(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{\mathrm{GL}_2}))_{\mathbb{Q}}$:

$$5\,[c_1(\mathcal{F}_{\sigma_\psi})] \;=\; 6\,[c_2(\mathcal{F}_{\sigma_\psi})] \tag{2}$$

between the Beilinson-regulator K-theory classes $[c_1(\mathcal{F}_{\sigma_\psi})]$ and $[c_2(\mathcal{F}_{\sigma_\psi})]$.

Remark 3.2 (Status and probability)**.** Conjecture 3.1 is speculative; we estimate a prior probability of 30–40%. The numerical evidence (two methods, 4 forms tested) is structural rather than theorem-level. The categorical identity (2) requires: (i) a precise definition of the Beilinson regulator map into $K_0(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{\mathrm{GL}_2}))_{\mathbb{Q}}$ in the number-field setting (currently undefined in [15]); and (ii) verification that the weight/twist normalization is compatible with the Damerell–Shimura period conventions.

Remark 3.3 (Why this is genuinely new)**.** The finding $R(f) = 6/5$ and its Ω-independence are not extracted from any prior literature. The explicit CM specialization at $\tau = i$ (lemniscate point) provides a concrete arithmetic instance where the abstract categorical framework of [6] speaks. We are unaware of any prior occurrence of the value $6/5$ in the geometric Langlands literature for GL_2 at CM points.

4. Why this is a categorical lift

Geometric Langlands over curves. In 2024–2025, Gaitsgory, Raskin, and collaborators (Arinkin, Beraldo, Campbell, Chen, Faergeman, Lin, Rozenblyum) completed the proof of the geometric Langlands conjecture (GLC) for GL_n over curves over $\mathbb{C}$ in five papers [6, 7, 8, 9, 10]. The key categorical structure on the spectral side is the ∞-category $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{\mathrm{GL}_n})$ of ind-coherent sheaves with nilpotent singular support on the stack of GL_n-local systems. For $n = 2$, the relevant K-theory group is $K_0(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{\mathrm{GL}_2}))_{\mathbb{Q}}$.

Beilinson's regulator (see [1]) sends motivic cohomology classes $[Z] \in H^{j+1}_{\mathcal{M}}(M_f, \mathbb{Q}(j+1))$ to periods $\langle [Z], \omega_f \rangle \in \mathbb{C}$ via the Beilinson–Deligne pairing. For $j = 1, 2$ these pairings recover (up to explicit factors) the special values $L(f, 1)$ and $L(f, 2)$ modulo the Shimura period Ω_f, by the conjectures of Beilinson as verified in special cases by Brunault, Kings, Loeffler, and Zerbes (see [12] for the Rankin–Selberg context).

The ratio $R(f) = \pi \cdot L(f, 1)/L(f, 2)$ is then the ratio of two such regulator pairings, stripped of Ω_f. The conjecture that $R(f) = 6/5$ lifts to an identity in K_0 is the assertion that the two regulator classes $[c_1]$ and $[c_2]$ satisfy a simple integral relation (2) under the spectral-side identification furnished by GLC.

Arithmetic-to-geometric transfer. The transfer from the arithmetic setting (automorphic form over $\mathbb{Q}$) to the geometric setting (sheaf on moduli of local systems) is itself the central open problem. Scholze's Bourbaki lecture 1252 (March 2026) [15] proposes Conjecture 1.5, a "template for a conjecture" extending GLC to number fields via "Gestalts of Langlands parameters," with the explicit warning: *"genuinely new ideas are still needed in the global number field case."* The arithmetic-to-geometric interface is further surveyed in Zhu's 2025 ICM preprint [17].

The four-layer M55 uniqueness analysis (conductor minimality, Steinberg edge $\varepsilon_2 = -1$, twist obstruction, Ω-independent ladder rationality) precisely characterizes **4.5.b.a** within the class of CM weight-5 forms over $\mathbb{Q}$, and the Conjecture 3.1 translation table is:

Classical (automorphic)	Spectral (Gestalt, Scholze Conj. 1.5)
$K = \mathbb{Q}(i)$, discriminant -4	dihedral image, restricted-from-$\mathbb{Q}(i)$
Conductor $(1+i)^2$	Artin conductor at $p = 2$, minimal
Steinberg $\varepsilon_2 = -1$	Frobenius semisimple at $p = 2$, eigenvalue -4
$R(f) = 6/5 \in \mathbb{Q}$	K_0 identity (2)

This translation is a restatement, not a new theorem.

5. Falsifier protocol

Stage 1 (completed, 2026-05-06). *SageMath cross-check at* 1.0×10^{-16}. Using `Newforms(Gamma1(4),5)` to identify f via $a_2 = -4$ and the built-in `f.lseries()` engine, one obtains $|R(f) - 6/5| < 1.1 \times 10^{-16}$ at Sage's default 53-bit precision. This constitutes an independent verification of the PARI result. **Status: PASS.** Higher-precision Sage verification (e.g. 10^{-30}) would require either increased `f.lseries().num_coeffs(N)` or an interface to LFunctions package with explicit precision.

Stage 2 (open, estimated 50–100 CPU-hours). *Eisenstein-symbol regulator verification.* Using the Kato–Loeffler–Zerbes (KLZ) Eisenstein-symbol machinery [12], construct explicit classes $\xi_j^{\mathrm{Eis}} \in H^2_{\mathcal{M}}(M_f, \mathbb{Q}(j+1))$ for $j = 1, 2$ in the motivic cohomology of the motive M_f, via modular units. Compute the Beilinson–Deligne pairings

$$\mathrm{reg}_j \;:=\; \langle \xi_j^{\mathrm{Eis}}, \omega_f \rangle_{\mathrm{BDS}} \;/\; \Omega_f$$

numerically (see Brunault [2] for relevant explicit regulator computations) and verify:

$$\frac{\mathrm{reg}_1}{\mathrm{reg}_2} \;\stackrel{?}{=}\; \frac{6}{5}$$

to 30-digit precision.

Explicit falsifier: any deviation $|\mathrm{reg}_1/\mathrm{reg}_2 - 6/5| > 10^{-20}$ at 30-digit verified precision kills Conjecture R3-C-1.

Collaborators required: Loeffler–Zerbes (for KLZ code on CM forms), Brunault (for Beilinson–Deligne explicit pairing). Outreach recommended before Stage 2 execution.

6. Bibliography

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